

Fixed Gaps

A Mathematical Theory of Relational Invariants

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Preface

This book began as a single observation: the gap between one person and another — an age difference, a blood relation — is fixed at the moment it becomes defined, and never changes afterward. The chapters that follow take that observation as far as mathematics will carry it.

Each chapter was written as a standalone document and resolves the open problems posed by its predecessor. Cross-references of the form “theory.md §2” refer to the chapters by their original file names, in order: *The Essay* (essay.md), *A Mathematical Theory of Fixed Gaps* (theory.md), *Readings over a Fixed-Gap Substrate* (readings.md), *The Frontier* (frontier.md), *Forces, Black Swans, and the Privilege of the Line* (forces.md), and *Melting, Frustration, and the Causal Web* (finale.md). A seventh chapter (application.md) answers an external review’s request for a worked application, deriving and numerically verifying three laws of rating systems; an appendix records the review and the response.

The arc, compressed: what persists between things are relations, not conditions; gaps are fixed exactly when everything drifts together; what we experience as change is the drift of non-invariant readings over an invariant substrate; which gaps survive depends on the symmetry group and the noise; order is the privilege of a one-dimensional time; and in the deepest description we have, the world does not have a gap structure — it is one.

The Essay

An essay on relational invariants — the things between us that never change.

1. The observation

Between any two people, certain quantities are frozen at the moment they first become defined, and no amount of time, effort, or reinterpretation can alter them. If someone is twenty years older than you, they are twenty years older than you forever. If someone is your sister, she is your sister forever. These are not properties of either person alone; they are properties of the *pair* — and unlike almost everything else about a pair of people, they do not move.

Call these **fixed gaps**: relations between two entities that are invariant over time.

2. The gap is fixed; its meaning is not

A twenty-year age gap is an unbridgeable chasm between a five-year-old and a twenty-five-year-old. Between a fifty-year-old and a seventy-year-old, it is nearly nothing. The absolute difference never changed — but the *ratio* of the two ages shrinks toward one as both grow.

This single arithmetic fact explains a great deal of family life. Parents and children can eventually become something like friends not because the gap closed, but because the proportion did. What feels like “growing closer” to an older relative is often just this ratio effect playing out on schedule.

The general principle: **the invariant is the substrate; the meaning is a non-invariant function of it**. What changes over a lifetime is never the gap itself, only the weight and reading we give it.

3. Fixed relations remove negotiation

Friendships, marriages, and collegueships are chosen — and therefore revocable, and therefore in need of maintenance. Blood relations and age gaps are given, not chosen. That is why they can bear weight that chosen relationships cannot: you cannot be fired as a brother. It is also why they can trap: there is no exit clause, only distance.

And because the structural facts between two people are frozen, everything that *does* change — trust, resentment, intimacy — must live entirely in interpretation and conduct. Family conflicts are rarely about the fixed facts; they are fights over what the fixed facts should be taken to mean.

4. Positions rotate through a fixed lattice

The age structure of a family is rigid, so people do not change their relations — they change their *location in* them. You do not stop being a child; you additionally become a parent, then a grandparent. Much of what we call maturing is discovering, from the other side, what a relation you have been inside your whole life actually felt like to the other person.

5. Beyond people: gaps between objects

The pattern is not human at all. Any two things that came into existence carry a fixed formation gap forever — two stars, two rocks, two documents, two blocks in a chain. Geology and archaeology are built on this: strata do not give absolute dates directly; they give *order* and *intervals*, and the intervals are incorruptible. Causal order is stronger still: if A caused B, no reinterpretation ever reverses it.

6. In physics, often only the gaps exist

This is the deepest version of the observation. Potential energy has no absolute value — only energy *differences* are physical. Voltage is a potential *difference*; the zero point is a convention. Position has no absolute meaning; only distances and relative velocities are measurable.

So it is not merely that gaps between objects stay fixed while the objects change. For many physical quantities, **the gap is the only measurable reality**, and the “absolute value” was a fiction penciled in for convenience. (The theory document makes this precise: every gap structure admits an absolute-value assignment, but only up to an arbitrary constant — the gauge freedom is exactly the fictional part.)

7. The relativity caveat — and why it strengthens the thesis

Einstein showed that some gaps we assumed were fixed are not: simultaneity, elapsed time, and length all depend on the observer. But physics responded by finding the *truly* invariant gaps underneath — the spacetime interval, proper time, causal order. When one inventory of fixed gaps dissolved, the framework was rebuilt around a deeper inventory. The instinct that reality is anchored in invariant relations survived; only the list of which relations qualify was revised.

The lesson generalizes: **which gaps are fixed depends on which transformations you admit**. Enlarge the group of admissible transformations and fewer gaps survive — but the survivors are more fundamental. This is Felix Klein’s Erlangen program applied to pairs of things.

8. The philosophical payoff: structuralism

If the changeable stuff (states, appearances, interpretations) is the surface, and the fixed gaps are the skeleton, one is led to a position philosophers call structuralism or relationalism — Leibniz’s side of his dispute with Newton. Objects may not have intrinsic, standalone natures at all; a thing may be nothing more than the bundle of its fixed relations to everything else.

The theory document turns this into a small theorem: the complete table of gaps determines the entire configuration of objects uniquely, up to a single global translation — that is, up to precisely the information (an absolute origin) that was never real in the first place.

9. Compressed

- Between people and between objects alike, what persists are **relations, not conditions**.
- The more fundamental the domain, the more the relations turn out to be **all there is**.
- What we experience as change is the drift of non-invariant readings over an invariant substrate — ratios shrinking, positions rotating through a fixed lattice, meanings renegotiated over facts that cannot be.

The mathematics is developed in theory.md.

A Mathematical Theory of Fixed Gaps

Companion to essay.md. We formalize “fixed gaps” — relations between entities that are invariant over time — and prove the essay’s claims.

The theory has three pillars:

1. **Gap structures** (§1–§4): quantitative gaps as torsor-like structures, where only differences are defined. Main results: absolute values exist only up to gauge (Prop 1.3), gaps are fixed *iff* everything drifts together (Thm 2.3), the gap table determines the world up to translation (Thm 3.1), and fixed additive gaps fade in relative terms (Thm 4.1).
2. **Gaps as group invariants** (§5): the Klein-program view. Which gaps are fixed depends on which transformations you admit; relativity is a change of group, not a refutation.
3. **Append-only worlds** (§6): qualitative gaps like kinship and causal order. Main result: positive existential facts are exactly the kind that, once true, stay true (Prop 6.2).

Throughout, A is a nonempty set of *objects* and $(G, +)$ an abelian group of *gap values* (typically $\mathbb{R}$; §8 drops commutativity).

1. Gap structures

Definition 1.1 (gap structure). A G -gap structure on A is a map $\delta : A \times A \rightarrow G$ satisfying the **cocycle identity**

$$\delta(a, c) = \delta(a, b) + \delta(b, c) \quad \text{for all } a, b, c \in A.$$

It is *separated* if $\delta(a, b) = 0 \implies a = b$.

The cocycle identity is the essence of gap-ness: gaps compose along chains. The age gap from you to your grandmother is your gap to your mother plus hers to her mother.

Lemma 1.2. $\delta(a, a) = 0$ and $\delta(b, a) = -\delta(a, b)$.

Proof. Setting $a = b = c$ in the cocycle identity gives $\delta(a, a) = 2\delta(a, a)$, so $\delta(a, a) = 0$. Then $\delta(a, b) + \delta(b, a) = \delta(a, a) = 0$. ■

Proposition 1.3 (gauge freedom — “the zero point is a convention”). Fix any $a_0 \in A$ and define $f(x) := \delta(a_0, x)$. Then

$$\delta(a, b) = f(b) - f(a) \quad \text{for all } a, b.$$

Moreover, if f' is any other function with $\delta(a, b) = f'(b) - f'(a)$, then $f' - f$ is constant. Conversely, every $f : A \rightarrow G$ defines a gap structure by $\delta(a, b) := f(b) - f(a)$.

Proof. $f(b) - f(a) = \delta(a_0, b) - \delta(a_0, a) = \delta(a, a_0) + \delta(a_0, b) = \delta(a, b)$ by the cocycle identity and Lemma 1.2. If $f'(b) - f'(a) = f(b) - f(a)$ for all a, b , then $(f' - f)(b) = (f' - f)(a)$, i.e. $f' - f$ is constant. The converse is a direct check of the cocycle identity. ■

Interpretation. Gap structures on A are exactly *potentials modulo constants*. You can always assign absolute values (a birthdate coordinate, a voltage, a height above sea level) consistent with the gaps — but never canonically: the assignment is unique only up to a global constant. The “absolute value” is real *as a gauge choice* and fictional *as a fact*. This is precisely why the zero of potential energy, of voltage, and of longitude are conventions, while their differences are physics.

Note that δ is separated iff f is injective, so a separated gap structure embeds A into G ; if f is also surjective, A is a G -torsor — a copy of the group that has forgotten its identity element. Time itself is the paradigm: instants form an $\mathbb{R}$ -torsor; there is no “instant zero,” only durations.

2. Dynamics: when is a gap fixed?

Definition 2.1. An *evolution* of A is a family of maps $\varphi_t : A \rightarrow A$, $t \in T$ (time; any monoid acting on A). The gap structure δ is **fixed under** φ if

$$\delta(\varphi_t a, \varphi_t b) = \delta(a, b) \quad \text{for all } a, b \in A, t \in T.$$

Proposition 2.2 (common drift $\Rightarrow$ fixed). Suppose the evolution is a *uniform drift*: there exist $g_t \in G$ with $f(\varphi_t a) = f(a) + g_t$ for all a (every object’s potential advances by the same amount). Then δ is fixed under φ .

Proof. $\delta(\varphi_t a, \varphi_t b) = f(\varphi_t b) - f(\varphi_t a) = (f(b) + g_t) - (f(a) + g_t) = \delta(a, b)$. ■

Ages are the motivating case: with birthdates b_i and $\$f = \$ \text{ age}$, $f_t(i) = t - b_i$ advances at rate 1 for everyone, so all age gaps are conserved.

The converse is the substantive theorem: uniform drift is not just *a* way to keep gaps fixed — it is the *only* way.

Theorem 2.3 (rigidity). Let δ be a separated gap structure and $\varphi : A \rightarrow A$ any map with $\delta(\varphi a, \varphi b) = \delta(a, b)$ for all a, b . Then $h(a) := \delta(a, \varphi a)$ is constant in a — i.e. φ is a uniform drift.

Proof. For any a, b :

$$\delta(a, \varphi a) = \delta(a, b) + \delta(b, \varphi b) + \delta(\varphi b, \varphi a) = \delta(a, b) + \delta(b, \varphi b) - \delta(\varphi a, \varphi b) = \delta(b, \varphi b),$$

using the cocycle identity, Lemma 1.2, and gap-preservation. ■

Corollary 2.4. The symmetry group of a separated gap structure — all invertible gap-preserving self-maps — is exactly the group of uniform drifts, a subgroup of G acting by translation. For a torsor it is all of G .

Interpretation. Fixed gaps and lockstep motion are *equivalent*. Age gaps are frozen because everyone ages at the same rate; Theorem 2.3 says any frozen quantitative gap forces an “equal aging rate” for the quantity behind it. Conversely, gaps break exactly when drift stops being uniform — which is precisely what special relativity discovered about elapsed time (§5). This is a small Noether principle: the conserved quantity (all pairwise differences) corresponds exactly to the symmetry (global translation).

3. Reconstruction: structuralism as a theorem

Theorem 3.1. Let δ be a separated G -gap structure on A . Then the potential $f : A \hookrightarrow G$ of Prop 1.3 is determined by the gap data uniquely up to a single additive constant. Consequently:

1. The full gap table $\{\delta(a, b)\}_{a, b \in A}$ determines the configuration of all objects up to one global translation of G .
2. Any two gap structures with the same gap table are isomorphic via a unique gap-preserving bijection matching corresponding objects.

Proof. Immediate from Prop 1.3 (existence and uniqueness-up-to-constant of potentials) and Theorem 2.3 (gap-preserving maps are translations). ■

Interpretation. The relations *are* the reality: the complete table of gaps pins down everything about the configuration except an absolute origin — which is exactly the datum that was never observable in the first place. This is the essay’s structuralist claim (Leibniz contra Newton) in theorem form. Nothing about the objects’ “positions” exceeds the gap data plus one unphysical choice.

4. The convergence theorem: why fixed gaps fade

Additive invariance and multiplicative invariance behave oppositely, and the human phenomenology of age gaps falls out of the additive case.

Theorem 4.1 (relative fading). Let $x(t), y(t) \rightarrow \infty$ with $x(t) - y(t)$ bounded. Then $x(t)/y(t) \rightarrow 1$.

Proof. $x/y = 1 + (x - y)/y$, and the numerator is bounded while $y \rightarrow \infty$. ■

Corollary 4.2 (ages). For ages $a_i(t) = t - b_i$, every pairwise gap $a_i - a_j = b_j - b_i$ is exactly conserved, while every ratio $a_i/a_j \rightarrow 1$. All beings become contemporaries in the limit.

Contrast 4.3 (multiplicative gaps diverge). If instead the *ratio* is invariant, $x(t) = \lambda y(t)$ with $\lambda > 1$ and $y \rightarrow \infty$, then $x - y = (\lambda - 1)y \rightarrow \infty$: fixed relative gap, exploding absolute gap. Compound growth (wealth, populations, citation counts) lives in this regime.

Interpretation. “Which kind of gap does the domain conserve?” is the first question to ask of any inequality. Domains with additive conservation (age, seniority) *equalize in proportion* over time; domains with multiplicative conservation (compounding capital) *diverge in absolute terms* while looking statically “fair” in ratio. The lived softening of the parent–child hierarchy and the widening of compounding wealth are the two branches of the same dichotomy.

5. Gaps as invariants of a group: the Klein view

Sections 1–4 assumed the gap values were differences in an abelian group. The general notion drops that: a gap is *anything about a pair that the admissible transformations cannot touch*.

Definition 5.1. Let a group Γ act on a state space S . A Γ -gap with values in a set V is a map $g : S \times S \rightarrow V$ invariant under the diagonal action:

$$g(\gamma x, \gamma y) = g(x, y) \quad \text{for all } \gamma \in \Gamma, x, y \in S.$$

Write $\text{Gaps}(\Gamma)$ for the set of such invariants.

Proposition 5.2 (more symmetry, fewer gaps). If $\Gamma \subseteq \Gamma'$ then $\text{Gaps}(\Gamma') \subseteq \text{Gaps}(\Gamma)$.

Proof. Invariance under every element of Γ' includes invariance under every element of Γ . ■

Examples.

Admissible group Γ	Surviving gaps	Casualties
time translations	age differences, order of birth	absolute dates
Euclidean isometries	distances, angles	absolute positions, orientations
Galilean group	relative velocities; distances at a common instant	absolute velocity (“the ether”)
Poincaré group	spacetime interval, proper time, causal order	simultaneity, elapsed coordinate time, spatial distance

Interpretation. “Is this gap fixed?” is not absolute — it is relative to the group of transformations the world (or the observer) is allowed to apply. Special relativity did not refute the fixed-gap thesis; it *enlarged* Γ , which by Prop 5.2 shrank the inventory of invariant gaps, leaving the deeper ones (the interval, the causal order). The research program of physics can be read as: find nature’s true Γ , then compute $\text{Gaps}(\Gamma)$ — and Theorem 2.3 is the special case saying that for one-dimensional additive quantities, conserving all pairwise gaps *is* translation symmetry.

6. Append-only worlds: kinship, causality, ledgers

Kinship gaps (“she is my sister”) are not numbers, and their fixedness has a different mechanism: not lockstep drift, but the *append-only* character of the past. We model it logically.

Definition 6.1. Fix a relational signature σ (e.g. binary $\text{parent}(x, y)$). A *history* is a chain of σ -structures

$$M_0 \subseteq M_1 \subseteq M_2 \subseteq \dots$$

where each inclusion is *append-only*: the universe may gain elements and relations may gain tuples, but no element or atomic fact is ever removed. A property $P(\bar{a})$ is **stable** (a fixed gap) if $M_s \models P(\bar{a})$ implies $M_t \models P(\bar{a})$ for all $t \geq s$.

Proposition 6.2. Every *existential-positive* formula — built from atomic formulas using only $\wedge$, $\vee$, and $\exists$ — is stable. Formulas involving negation or universal quantification need not be.

Proof. Induction on formula structure. Atomic facts persist by the append-only condition. Conjunction and disjunction preserve persistence. For $\exists x \psi$: a witness in M_s still exists and still satisfies ψ in M_t by the inductive hypothesis. For the negative claim, two counterexamples: $\neg \exists y \text{parent}(x, y)$ (“ x is childless”) is destroyed by appending a child; $\forall y (\text{parent}(x, y) \rightarrow \text{parent}(z, y))$ is destroyed by appending a child of x alone. ■

Application (kinship). Parenthood edges are appended at birth and never deleted. Hence *sibling* ($\exists z \text{parent}(z, x) \wedge \text{parent}(z, y)$), *grandparent*, *cousin*, and *ancestor* (a monotone transitive closure, an increasing union of existential-positive formulas) are all stable — fixed forever once true. The asymmetry of family life falls out of the syntax: **positive kinship facts are fixed; negative ones (“childless,” “unmarried,” “only child”) are not.** Blood relations are fixed gaps not because of biology per se, but because the world never un-happens an event.

Application (causality and ledgers). Causal precedence is a partial order to which history only ever *adds* comparable pairs; an existing pair $a \prec b$ is never reversed — order facts are atomic, hence stable. A blockchain is this proposition turned into engineering: an append-only structure whose entire security model is making retro-deletion of facts computationally infeasible. It is a machine for *manufacturing* fixed gaps (confirmed precedence of transactions) in an adversarial environment.

Remark. Prop 6.2 is the finger-exercise version of preservation theorems in model theory (Łoś–Tarski; the homomorphism-preservation theorem of Rossman), which characterize stable properties as *exactly* the existential-positive ones under suitable hypotheses. In slogan form: *the fixed gaps of an append-only world are precisely its positive existential facts.*

7. The meaning layer: readings over a frozen substrate

The essay claims: “the gap is fixed, but its meaning is not.” Formally:

Definition 7.1. Given a gap structure with potential f_t evolving by uniform drift (so all gaps are fixed), a *reading* is any function $m(a, b, t)$ of the pair’s full state — not required to be invariant.

Readings vary even though gaps do not, because they can depend on the absolute potentials, not just their difference. Three canonical readings of an age pair with fixed gap $d = f(b) - f(a) > 0$:

- **Ratio reading** $m = f_t(b)/f_t(a)$: decays monotonically to 1 (Thm 4.1). *Authority, era-membership, “different generations.”* Fades on schedule.
- **Threshold reading** $m = \mathbf{1}[f_t(a) \geq \theta]$: flips once, permanently, at a computable time. *Adulthood, retirement, outliving.*
- **Positional reading** $m = \$$ the pair’s location in a fixed order lattice: never changes, but each object’s *other* relations accumulate around it (§6 appends). *Becoming a parent while remaining a child.*

Claim (informal). The dynamics of long-term relationships is exhausted by the motion of readings over the frozen gap substrate. Nothing structural moves; everything interpretive does — and the three reading types above (decaying ratios, one-way thresholds, accumulating positions) are the three characteristic time-signatures of that motion.

8. Extensions and open problems

1. **Non-abelian gaps.** Replace $(G, +)$ with any group and the cocycle identity with $\delta(a, c) = \delta(a, b) \cdot \delta(b, c)$. Everything in §1–§3 survives with care about sides. Example: relative orientation between rigid bodies ($G = SO(3)$) — a fixed gap when bodies rotate in lockstep.
2. **Categorical form.** A gap structure is exactly a functor from the pair groupoid of A (objects A , one isomorphism between each pair) to G viewed as a one-object groupoid; the cocycle identity is functoriality. Fixedness of δ under an evolution is naturality. What do gap structures valued in a general groupoid buy?
3. **Noisy gaps.** Gaps fixed only in expectation, or as martingales: $\mathbb{E}[\delta_{t+1} \mid \mathcal{F}_t] = \delta_t$. Which of Theorems 2.3 and 3.1 survive in distribution? (Candidate domain: relative skill ratings, exchange rates.)
4. **Almost-fixed gaps.** Slowly varying invariants ($|\dot{\delta}| \leq \epsilon$, adiabatic invariants): quantify how much of the reconstruction theorem (Thm 3.1) degrades with ϵ — a stability-of-structuralism question.
5. **n -ary gaps.** Invariants of triples and n -tuples under the diagonal action (Def 5.1 with S^n): birth-order permutations, angles, cross-ratios. The cross-ratio — the fixed gap of projective geometry — is a 4-ary example, suggesting a hierarchy: which geometries have no fixed binary gaps at all, only higher-arity ones? **Resolved in frontier.md §3:** the hierarchy is the sharp-transitivity ladder — a sharply k -transitive group has its first invariant at arity $k + 1$ (gap, ratio, cross-ratio at $k = 1, 2, 3$) — and by Tits’s

theorem on sharply 4-transitive groups the ladder terminates at the cross-ratio, leaving order type as the sole invariant of maximal symmetry.

6. **Classifying readings.** Given a fixed-gap substrate, classify all readings by asymptotic behavior. Conjecture (informal): under mild regularity, every reading of an additively drifting pair decomposes into the three time-signatures of §7 — decaying, one-way switching, and accumulating components. **Resolved in readings.md:** false as a trichotomy — there is an unavoidable fourth signature, *recurrent* readings (anniversary cycles) — but true as an amended tetrachotomy. The three conjectured types are recovered as the building blocks of all locally-BV readings, the signature is computable from the reading’s symmetry (with a quantitative *hyperbolic fading law*: gap memory decays as d/t), and the recurrent class factors through a new fixed gap valued in a quotient group (the *phase gap*), strengthening the structuralist thesis.

Readings over a Fixed-Gap Substrate

Resolution of Conjecture 8.6 of theory.md.

Conjecture 8.6 (restated). Under mild regularity, every reading of an additively drifting pair decomposes into three time-signatures: decaying, one-way switching, and accumulating.

Verdict. *False as stated, true as amended.* There is an unavoidable fourth signature — **recurrent** readings (anniversary cycles) — exhibited by an explicit counterexample (§4). But the amended classification is a theorem (§5), and the fourth class strengthens rather than weakens the structuralist thesis: every periodic reading factors through a *new fixed gap* valued in a quotient group — the **phase gap** (§4.3). What looks like endless oscillation is frozen structure in a smaller group.

Two byproducts are of independent interest: the **hyperbolic fading law** (Thm B: the felt size of a fixed gap decays exactly like d/t , with a universal constant), and the **pushforward lemma** (Lem 4.2: homomorphisms manufacture new fixed gaps from old).

1. Setup and the drift-derivative lemma

As in theory.md §2, take a pair with fixed gap $d > 0$ under uniform additive drift. Normalizing the gauge (Prop 1.3 there), the pair's state at time t is

$$s(t) = (x(t), x(t) + d), \quad x(t) = t,$$

a point moving along a line of slope 1 in the potential plane $\{(u, v)\}$. A **reading** is $m(t) = F(s(t))$ for some $F : \Omega \subseteq \mathbb{R}^2 \rightarrow \mathbb{R}$ defined on the reachable states. The four **time-signatures** of a reading:

- **frozen** — m constant;
- **decaying** — $m(t)$ converges as $t \rightarrow \infty$ (but is not constant);
- **accumulating** — $m(t) \rightarrow +\infty$ or $-\infty$;
- **recurrent** — $\liminf m < \limsup m$: the reading revisits values forever.

These four are trivially exhaustive and mutually exclusive; the content of a classification is in *computing* the signature from the structure of F , and in identifying the building blocks of general readings. That is what follows.

Lemma 1.1 (drift derivative). If F is differentiable along the flow direction, then

$$\dot{m}(t) = (\mathcal{L}F)(s(t)), \quad \mathcal{L} := \partial_u + \partial_v,$$

the derivative of F along the drift. Consequently m is frozen for every pair and every time iff $\mathcal{L}F \equiv 0$ on Ω , and the signature of m is the convergence class of $\int^t (\mathcal{L}F)(s(\tau)) d\tau$.

Proof. Chain rule along $s(t) = (t, t + d)$; the integral statement is the fundamental theorem of calculus. ■

Proposition 1.2 (frozen = gap: closing the Klein circle). m is frozen for every pair iff F is invariant under the drift group (translations along $(1, 1)$), i.e. iff $F(u, v) = G(v - u)$ for some G — iff F is a function of the gap alone.

Proof. Frozen for every pair means F is constant on every line of slope 1 in Ω ; these lines are exactly the level sets of $v - u$. (With differentiability, this is the method of characteristics for $\mathcal{L}F = 0$; without it, the argument above is direct.) ■

This closes a circle left open in theory.md: there, gaps were *defined* structurally (§1) and characterized by symmetry (§5); here they reappear dynamically — **the fixed gaps are exactly the frozen readings**. Definition 5.1’s invariants and the constant observables of the flow are the same class.

2. The symmetry dictionary

The signature of a reading is computable from its behavior under two groups acting on the potential plane: translations (the drift) and dilations. The dictionary:

F invariant under / homogeneous of	signature	rate
translations along the drift	frozen	—
dilations, degree 0	decaying	gap memory $\sim d/t$
dilations, degree $k > 0$	accumulating	$\sim t^k$; gap relatively negligible

Theorem A (translation-invariant readings). Covered by Prop 1.2: they are frozen, and they are precisely the gap functions.

Theorem B (hyperbolic fading law). Let F be positively homogeneous of degree 0 (scale-invariant: $F(\lambda u, \lambda v) = F(u, v)$ for $\lambda > 0$) and C^1 near $(1, 1)$. Then for every gap d :

$$m(t) = F(1, 1) + F_v(1, 1) \frac{d}{t} + O\left(\frac{d^2}{t^2}\right).$$

In particular:

1. $m(t) \rightarrow F(1, 1)$ — a **universal limit**, independent of the gap: all pairs read the same in the end.
2. The memory of the gap decays hyperbolically, as d/t , with a constant $F_v(1, 1)$ depending on the reading but not the pair.

Proof. By degree-0 homogeneity, $F(t, t + d) = F(1, 1 + d/t)$. Taylor expansion at $(1, 1)$ gives the formula. (By Euler's relation $uF_u + vF_v = 0$, at $(1, 1)$ we have $F_u = -F_v$, so one derivative controls everything.) ■

Example. The ratio reading $F(u, v) = v/u$ gives $m(t) = (t + d)/t = 1 + d/t$ exactly: the felt seniority of an elder is d/t , i.e. Theorem 4.1 of theory.md with an explicit rate. The “generation gap” does not fade linearly or exponentially — it fades **hyperbolically**: fast at first, then ever more slowly, never quite gone.

Corollary B' (order rigidity of felt gaps). To first order, two pairs with gaps d and d' have felt-gap ratio d/d' at *every* time. Fading is uniform across pairs: the ordering of relationships by felt gap compresses toward equality but **never inverts**. (Your two elders always feel unequal in the same order; they just both fade.)

Theorem C (homogeneity dictionary and drift duality).

1. If F is positively homogeneous of degree $k > 0$, continuous at $(1, 1)$, with $F(1, 1) \neq 0$, then $m(t) = t^k F(1, 1 + d/t) \sim t^k F(1, 1)$: **accumulating**, with the gap entering only at relative order d/t .
2. **Duality.** Under *exponential* drift ($x(t) = x_0 e^t$, multiplicative gap $v/u = \rho$ fixed — the compounding regime of theory.md §4.3), the roles swap: scale-invariant readings are frozen ($F(x, \rho x) = F(1, \rho)$), and translation-invariant readings accumulate ($G(v - u) = G((\rho - 1)x_0 e^t)$, divergent for monotone unbounded G).

Proof. (1) is the homogeneity identity plus continuity. (2) is direct substitution; the two regimes are conjugate under exp, which maps additive drift to multiplicative drift and difference gaps to ratio gaps. ■

Interpretation. Theorem C(2) generalizes the additive/multiplicative dichotomy of theory.md §4 into a single principle: **a reading is frozen iff it is invariant under the drift group** (Prop 1.2 in the additive case), and the interesting long-run behavior belongs to readings invariant under the *other* group. Age-world (additive drift): differences frozen, ratios decay. Capital-world (multiplicative drift): ratios frozen, differences explode. Same theorem, two one-parameter subgroups of the affine group.

3. Building blocks: the Jordan decomposition of readings

The symmetry dictionary handles structured readings. For *general* (tame) readings, the three conjectured signatures appear as building blocks.

Theorem D (decomposition of tame readings). Let $m : [t_0, \infty) \rightarrow \mathbb{R}$ be of locally bounded variation. Then:

1. $m = P - N$ where P, N are nondecreasing (two **accumulating-type** components: the running positive and negative variation).
2. $m = m_{\text{cont}} + \sum_i h_i \mathbf{1}[t \geq \theta_i]$, where the sum is a countable, locally finite superposition of **threshold readings** (the jump part) and m_{cont} is continuous; the decomposition is unique.
3. If the total variation of m is finite, then P and N are bounded monotone, hence convergent, hence m converges: **decaying** signature.

4. If m is eventually monotone with infinite total variation, then (since local variation is finite) the tail variation is infinite, so the monotone tail is unbounded: **accumulating** signature.

Proof. (1) is the Jordan decomposition. (2) is the standard splitting of a BV function into its continuous part and its (countable, locally summable) jumps. (3): bounded monotone functions converge. (4): a monotone function with infinite variation on $[T, \infty)$ is unbounded there. ■

So the conjecture's three types are exactly the atoms of tame readings: every locally-BV reading is *built from* thresholds and accumulating parts, and *converges* precisely when its total variation is finite. What Theorem D cannot rule out is a reading whose variation is infinite while it remains bounded and non-monotone — and that case is real:

4. The fourth signature: recurrence, and the gaps hiding inside it

4.1 Counterexample to the strict trichotomy. Take $F(u, v) = \sin(2\pi u/p)$: the reading “where is a in its yearly cycle” — a birthday clock. Then $m(t) = \sin(2\pi t/p)$ is bounded, non-convergent, non-monotone, with infinite total variation: **recurrent**, and not a locally finite superposition of the three conjectured types with a convergent remainder. Conjecture 8.6 as stated is false. Anniversaries are not decaying, not one-way, not accumulating — they *return*.

4.2 Lemma (pushforward of gaps). Let δ be a G -gap structure on A (theory.md Def 1.1) and $h : G \rightarrow H$ a group homomorphism. Then $h \circ \delta$ is an H -gap structure, and it is fixed under any evolution that fixes δ .

Proof. h preserves the cocycle identity because it preserves addition; fixedness is immediate. ■

Pushforwards manufacture derived fixed gaps: the parity of an age gap in whole years ($\mathbb{Z} \rightarrow \mathbb{Z}/2$), and — the case we need — the **phase gap** $\bar{d} := d \bmod p \in \mathbb{R}/p\mathbb{Z}$, the gap pushed into the circle group by $\mathbb{R} \rightarrow \mathbb{R}/p\mathbb{Z}$. The phase gap is the answer to “how far apart in the *calendar* are their birthdays?” — and it is as frozen as the gap itself.

4.3 Theorem E (recurrent readings factor through quotient gaps). Suppose F is *jointly p -periodic along the drift*: $F(u + p, v + p) = F(u, v)$. Then:

1. Every reading $m(t) = F(t, t + d)$ is p -periodic in t : pure recurrent signature (unless constant).
2. The orbit function $O_d : \mathbb{R}/p\mathbb{Z} \rightarrow \mathbb{R}$, $O_d(s) = F(s, s + d)$, is well defined, and $m(t) = O_d(t \bmod p)$.
3. Two pairs with gaps d, d' and the same birth phase have identical readings for all time iff $O_d = O_{d'}$; pairs with equal gap and different birth dates have identical readings *up to the fixed time shift* given by their phase gap.

Proof. (1) $m(t + p) = F(t + p, t + p + d) = F(t, t + d) = m(t)$. (2) Well-definedness is exactly joint p -periodicity. (3) is (2) applied to both pairs, with the shift read off from the pushforward gap of 4.2. ■

Interpretation. The recurrent class does not escape the fixed-gap framework — it *exposes more of it*. A periodic reading carries no new degrees of freedom beyond (i) a frozen orbit-shape determined by the gap and (ii) a frozen circle-valued phase. Whether two people’s birthdays fall in the same season, whether two commemorations collide, whether two cycles are in or out of phase: all frozen at inception, all pushforward gaps. **Anniversaries are how frozen structure stays perceptible:** a fixed gap is invisible in any single moment, but a recurrent reading parades it past the observer once per period, forever. Ritual is the phenomenology of the quotient torsor.

5. The amended classification theorem

Assembling §§1–4:

Theorem (Reading Classification). Let $m(t) = F(t, t + d)$ be a reading of a fixed-gap pair under uniform additive drift.

1. **(Frozen = invariant.)** m is frozen for every pair iff F is drift-invariant iff F is a function of the gap alone (Prop 1.2). The frozen readings are exactly the fixed gaps.
2. **(Symmetry determines signature.)** If F is homogeneous of degree 0 and C^1 near $(1, 1)$: decaying, with universal limit and hyperbolic gap memory $F_v(1, 1) d/t$ (Thm B). If homogeneous of degree $k > 0$ with $F(1, 1) \neq 0$: accumulating like t^k (Thm C). If jointly periodic along the drift: recurrent, factoring through the phase gap (Thm E).
3. **(Building blocks.)** Every locally-BV reading is a locally finite superposition of threshold readings plus a difference of two continuous accumulating components; it is decaying iff its total variation is finite, accumulating if eventually monotone with infinite variation, and recurrent otherwise (Thm D).
4. **(Exactly four.)** The signature classes {frozen, decaying, accumulating, recurrent} are exhaustive and mutually exclusive, and none is empty: the conjectured trichotomy misses precisely the recurrent class, which is nonempty (§4.1) but adds no unfrozen structure (Thm E).

Resolution of Conjecture 8.6: false as a trichotomy; true as the tetrachotomy above, with the conjectured three types recovered as the atoms of all tame readings (part 3) and the fourth type revealed as frozen structure in a quotient group (part 2, Thm E).

6. What this says about the essay’s claims

- “The gap is fixed but its meaning is not” now has a rate: for every smooth scale-invariant meaning, the felt gap is $c \cdot d/t$. Meaning decay is hyperbolic — steep in youth, asymptotically flat, never complete. And by Corollary B’, felt gaps compress but never reorder: time equalizes proportions while preserving every comparison.

- **The two worlds of §4 (theory.md) are one theorem:** what a domain freezes is the drift-invariant readings; what it lets run is the readings invariant under the conjugate group. Age-world freezes differences and bleeds ratios; capital-world freezes ratios and bleeds differences.
- **Ritual earns a mathematical role:** recurrent readings are the unique signature class that keeps a frozen gap *perceptible* indefinitely (frozen readings are imperceptible as change; decaying ones fade; accumulating ones drown the gap at relative order d/t). A culture that wants its fixed relations felt must institutionalize periodic readings — birthdays, memorials, festivals. The mathematics did not know it was going to say that, which is the best sign it is saying something.

7. What remains open

All three problems below are resolved in frontier.md.

1. **Beyond BV:** classify readings of bounded mean oscillation or almost periodic type; is every bounded recurrent reading asymptotically almost periodic under natural constraints on F ? **Resolved (no):** $\sin(\log t)$ is bounded, recurrent, and not asymptotically almost periodic — it is periodic under the *dilation* flow. Recurrence, like fixedness, is relative to a one-parameter group (frontier.md §1).
2. **Noisy drift:** with $x(t)$ a random walk with drift, which parts of the classification survive in distribution? (Conjecture: Thm B holds with d/t replaced by $d/\mathbb{E}x(t)$ plus a $O(t^{-1/2})$ fluctuation term.) **Resolved:** the conjectured form is correct, but the fluctuation term dominates: felt order survives only to a horizon $t^* \asymp (\Delta d)^2/\sigma^2$, the only noise-proof frozen readings are affine in the gap, and the present's information about a gap decays like $1/t$ unless records are kept (frontier.md §2).
3. **Multi-pair readings:** signatures of functions of n objects' states under the diagonal drift — where the cross-ratio and birth-order invariants of theory.md §8.5 should reappear as the frozen class. **Resolved:** n -ary frozen readings are exactly functions of the gap matrix; gap $\rightarrow$ ratio $\rightarrow$ cross-ratio is one construction at levels $k = 1, 2, 3$ of the sharp-transitivity ladder, which provably terminates (Tits), leaving birth order as the final invariant (frontier.md §3).

The Frontier: Slow Recurrence, Noise, and the Invariant Tower

Resolutions of the three open problems of readings.md §7, and of the n -ary question of theory.md §8.5.

Summary of verdicts:

- **Q1 (beyond BV).** *No*: bounded recurrent readings need not be asymptotically almost periodic — $\sin(\log t)$ is a counterexample (Thm 1.2). But the failure is structured: such readings are periodic under a *conjugate* flow (dilations), and every one-parameter symmetry group carries its own anniversary theory with its own frozen phase gaps (§1.3).
- **Q2 (noisy drift).** The conjectured form of the noisy fading law is correct — but the fluctuation term *dominates* the gap memory. Felt order between gaps survives only to a horizon **quadratic in the gap difference** (Thm 2.4), the only noise-proof readings are affine in the gap (Thm 2.5), and the present's information about a fixed gap decays like $1/t$ — unless records are kept (§2.6).
- **Q3 (n -ary invariants).** Gap, affine ratio, and cross-ratio are the same object at levels $k = 1, 2, 3$ of a single ladder (Thm 3.2), and by a theorem of Tits the ladder **provably terminates** at the cross-ratio (Thm 3.3). At maximal symmetry the only surviving invariant is birth order — the unkillable fixed gap (§3.4).

1. Slow recurrence: the answer to Q1 is no, and why that's good

1.1 The question. Readings.md classified tame readings and found four signatures, with the recurrent class generated (in the examples) by periodic structure. Q1 asked: is every *bounded recurrent* reading asymptotically almost periodic (a.a.p.) under natural constraints on F — i.e., is recurrence always calendar-like?

1.2 Theorem (counterexample). The reading $m(t) = \sin(\log t)$ — from $F(u, v) = \sin(\log u)$, continuous and bounded on the reachable states $t \geq t_0 > 0$ — is bounded and recurrent but *not* asymptotically almost periodic.

Proof. Recurrence: m attains ± 1 at $t = e^{\pi/2 + k\pi}$ for all large k . For a.a.p., we use the compactness criterion: m is a.a.p. iff its translate family $\{m(\cdot + s)\}_{s \geq 0}$ is relatively compact in $C_b([0, \infty))$ with the sup norm. Take $s_n = e^{2\pi n}$. On any compact $[0, K]$,

$$m(t + s_n) = \sin(2\pi n + \log(1 + t/s_n)) = \sin(\log(1 + t/s_n)) \longrightarrow \sin 0 = 0$$

uniformly. So any sup-norm limit g of a subsequence of translates vanishes on every compact, hence $g \equiv 0$; but every translate has sup norm 1 (the oscillation through ± 1 continues beyond any point), so no subsequence converges to 0 in sup norm. No convergent subsequence exists: m is not a.a.p. ■

1.3 Diagnosis: anniversaries under conjugate flows. The counterexample is not pathological — it is *periodic*, just not under the drift. $F(u, v) = \sin(\log u)$ satisfies $F(\lambda_0 u, \lambda_0 v) = F(u, v)$ for $\lambda_0 = e^{2\pi}$: it is periodic under the **dilation flow**. In additive time this appears as log-periodicity: recurrences at $t, \lambda_0 t, \lambda_0^2 t, \dots$ — return times growing geometrically. These are the **doubling anniversaries**: the 1st, 2nd, 4th, 8th year; the celebrations that thin out forever.

The general principle (a time-change lemma): a reading periodic under a one-parameter flow with clock $\theta(t)$ has the form $m(t) = P(\theta(t))$ with P periodic, hence is recurrent whenever $\theta(t) \rightarrow \infty$, and is almost periodic in t precisely when the clock is (asymptotically) proportional to t itself. Drift-periodic readings ($\theta = t$): calendar anniversaries, a.p. Dilation-periodic readings ($\theta = \log t$): slowing anniversaries, recurrent but never a.p. Every one-parameter subgroup of the affine group yields its own anniversary theory.

1.4 Phase gaps follow the drift, not the reading. The pushforward lemma (readings.md 4.2) applies in each regime *to that regime's own drift group*: in age-world (additive drift), calendar anniversaries carry the frozen phase gap $d \bmod p$ — two people's birthdays keep the same calendar distance forever. In capital-world (multiplicative drift), it is the *octave gap* $\rho \bmod \lambda_0^{\mathbb{Z}} \in \mathbb{R}_+/\lambda_0^{\mathbb{Z}}$ (a circle again, via the homomorphism $(\mathbb{R}_+, \times) \rightarrow \mathbb{R}_+/\lambda_0^{\mathbb{Z}}$) that is frozen. Mismatched pairs drift: in age-world, two people's *doubling* anniversaries come into and out of alignment, their coincidences thinning out forever, while their calendar birthdays stay locked. Which recurrences stay synchronized is itself a fixed-gap phenomenon — and it is decided by which group is doing the drifting.

Amended answer to Q1. Bounded recurrent readings are asymptotically almost periodic iff their recurrence is driven by the drift group itself (calendar-type); recurrence driven by conjugate flows is genuinely slower than any almost-periodic pattern. The classification of recurrent readings is *by which one-parameter group they are periodic under* — recurrence, like fixedness, is relative to a group.

2. Noisy drift: martingale gaps, the felt-order horizon, and records

2.1 The model. Discrete time keeps every proof elementary. Objects $i \in \{a, b, \dots\}$ with potentials

$$x_i(t+1) = x_i(t) + \mu_i(t) + w(t+1) + \xi_i(t+1),$$

where $\mu_i(t)$ is object i 's predictable drift rate, w is a **common** shock (the same for all objects; arbitrary, even heavy-tailed or adversarial), and ξ_i are **idiosyncratic** shocks: martingale differences, independent across objects, with variance σ^2 .

2.2 Proposition (noise dichotomy). Common noise, however wild, preserves every gap exactly: $\delta_{ab}(t) = x_b(t) - x_a(t)$ is unchanged by w . Under idiosyncratic noise, the gap satisfies

$$\delta_{ab}(t) = d + \sum_{s \leq t} (\mu_b - \mu_a)(s) + M(t),$$

with M a martingale of variance $2\sigma^2 t$; hence δ_{ab} is a martingale (**fixed in expectation**) iff $\mu_a \equiv \mu_b$.

Proof. The common term cancels in every difference — this is the rigidity theorem (theory.md 2.3) applied pathwise: common noise is still uniform drift, just a random one. The decomposition and the martingale criterion are immediate; the variance adds because the ξ_i are independent. ■

So the rigidity principle survives noise in sharpened form: *gaps are immune to any amount of shared turbulence; only asymmetric shocks move them, and equal mean drift keeps them fixed on average with $\sqrt{t}$ fluctuation.* Wars, recessions, pandemics — everything that hits both members of a pair alike — leaves every gap untouched.

2.3 The noisy fading law. For the ratio reading, with $\delta(t) = d + M(t)$ as above,

$$m(t) = \frac{x_b}{x_a} = 1 + \frac{d}{t} + \frac{M(t)}{t} + O_p(t^{-3/2}),$$

and the fluctuation term has typical size $\sqrt{2\sigma^2 t}/t = \sigma\sqrt{2/t}$. The conjectured form (readings.md §7.2) is thus correct — but note which term wins: the noise floor $\sigma\sqrt{2/t}$ **dominates** the gap memory d/t as $t \rightarrow \infty$. The signal-to-noise ratio of the felt gap is

$$\text{SNR}(t) = \frac{d/t}{\sigma\sqrt{2/t}} = \frac{d}{\sigma\sqrt{2t}} \rightarrow 0.$$

In the deterministic world the felt gap fades but remains exact; in the noisy world it fades *into the noise* — past a horizon, the gap is still there (in expectation) but can no longer be felt above the turbulence.

2.4 Theorem (felt-order horizon). Let two pairs have gaps $d > d'$, all four objects carrying independent idiosyncratic noise. The probability that their felt gaps are observed in inverted order at time t is

$$p(t) = \Phi\left(-\frac{d-d'}{2\sigma\sqrt{t}}\right) \nearrow \frac{1}{2},$$

(Gaussian shocks; in general by the CLT). Thus the order-rigidity corollary of readings.md (felt gaps compress but never reorder) survives noise only up to a **crossover horizon**

$$t^* \asymp \frac{(d-d')^2}{\sigma^2},$$

beyond which the comparison degenerates to a coin flip.

Proof. Inversion means $d + M_1(t) < d' + M_2(t)$ with M_1, M_2 independent, each of variance $2\sigma^2 t$; $M_2 - M_1$ has variance $4\sigma^2 t$, giving the stated probability, which is increasing in t with limit $1/2$. ■

The horizon is **quadratic in the gap difference**: relationships whose gaps differ by twice as much keep their felt order four times longer. This is the noisy world’s replacement for order rigidity — order is not preserved forever, but its half-life is computable.

2.5 Theorem (the noise-proof readings are affine). Let $\delta(t) = d + M(t)$ be a martingale gap with nondegenerate idiosyncratic noise. For a frozen reading $G(\delta)$ (frozen in the deterministic world by Prop 1.2 of readings.md):

1. If G is convex, $G(\delta(t))$ is a submartingale: $\mathbb{E} G(\delta(t))$ is nondecreasing, strictly increasing if G is strictly convex — **noise systematically inflates convex readings** even though the gap itself is fixed in expectation.
2. $\mathbb{E} G(\delta(t))$ is constant for all t (and all noise laws with full support) iff G is affine.

Proof. (1) is conditional Jensen plus the martingale property. (2): constancy for Gaussian noise forces G to be harmonic for the heat semigroup, i.e. affine; conversely affine G composed with a martingale is a martingale. ■

Deterministically, *every* function of the gap is frozen; noise collapses this algebra to its affine part. Any convex “cost of distance” — squared separation, penalties, worry — grows in expectation under symmetric noise. The gap doesn’t move on average; its convex readings do. This is a fixed-gap version of a familiar phenomenon: volatility is invisible to linear functionals and taxes all convex ones.

2.6 Epistemic fading, and what records are for. Observing only the present state, the gap estimate is $\delta(t) \sim \mathcal{N}(d, 2\sigma^2 t)$: the Fisher information about d carried by the present is $1/(2\sigma^2 t) \rightarrow 0$. **Under idiosyncratic noise, the present forgets the past at rate $\sqrt{t}$.** Ontologically the gap is still fixed in expectation; epistemically it dissolves — unless the birth values were *recorded*, in which case estimating d is reading, not inference.

This gives the append-only theme of theory.md §6 an epistemic twin, and completes a triptych:

- **Ontology:** append-only worlds make gaps fixed (facts are never deleted).
- **Phenomenology:** recurrent readings keep gaps *felt* (readings.md §6 — ritual).
- **Epistemology:** append-only records keep gaps *known* against the $1/t$ information decay (archives, genealogies, ledgers).

A civilization’s rituals and its archives are solutions to two different decay laws attacking the same invariants.

3. The invariant tower: gap, ratio, cross-ratio — and why it ends

3.1 n -ary frozen readings. For n objects under uniform additive drift, state $(x_1, \dots, x_n) = (t - b_1, \dots, t - b_n)$: a reading $F(x_1, \dots, x_n)$ is frozen for every configuration iff F is constant on the orbits of the diagonal translation action, iff F factors through the gap vector $(x_2 - x_1, \dots, x_n - x_1)$. (Same level-set argument as Prop 1.2 of readings.md.) The reconstruction theorem extends verbatim: the gap matrix of n objects determines the configuration up to one global translation.

3.2 Theorem (the first-invariant ladder). Let Γ act **sharply k -transitively** on a line-like space X (uniquely mapping any ordered k -tuple of distinct points to any other). Then:

1. There are no nonconstant Γ -invariants of j -tuples of distinct points for $j \leq k$.
2. The invariants of $(k + 1)$ -tuples are exactly the functions of a single **fundamental invariant**: the coordinate of the last point after the unique $\gamma \in \Gamma$ sending the first k points to a fixed standard frame.

Applied to the three classical actions on the line / projective line:

k	group Γ	standard frame	fundamental invariant of $(k+1)$ points
1	translations	$u \mapsto 0$	the gap $v - u$
2	affine $u \mapsto au + b$	$(u, v) \mapsto (0, 1)$	the ratio $\frac{w - u}{v - u}$
3	projective $PGL_2(\mathbb{R})$	$(u, v, w) \mapsto (0, 1, \infty)$	the cross-ratio $[u, v; w, z]$

Proof. (1): sharp k -transitivity makes distinct j -tuples ($j \leq k$) a single orbit, on which any invariant is constant. (2): the map “(tuple) $\mapsto$ image of the last point under the unique frame-fixing γ ” is well defined by sharpness, invariant by construction, and complete (two tuples with the same value are matched by composing the two frame maps). Computing it in each row gives gap, ratio, cross-ratio. ■

So the gap of theory.md §1, the ratio of the capital-world, and the cross-ratio of projective geometry are **one construction at three levels**: each is the first thing the symmetry group cannot destroy, appearing at arity $k + 1$ — one point beyond what the group can normalize away.

3.3 Theorem (the tower terminates). There is no sharply 4-transitive group acting on an infinite set (Tits, 1952; the sharply 4- and 5-transitive groups are the finite Mathieu groups M_{11} , M_{12} , plus small symmetric/alternating cases). Hence on the line the ladder of fundamental invariants is complete at three rungs:

$$\text{gap} \longrightarrow \text{ratio} \longrightarrow \text{cross-ratio} \longrightarrow \emptyset.$$

The cross-ratio is not just *a* higher invariant — it is provably the **last** of its kind, even allowing arbitrarily wild (discontinuous) group actions.

3.4 The far end: order, the unkillable invariant. Past the projective group lies the full order-automorphism group $\text{Homeo}_+(\mathbb{R})$ — every continuous, order-preserving reparameterization of the timeline, including clocks that run unevenly. It is k -transitive for every k (not sharply), so *all* numerical invariants die; what survives of an n -tuple is exactly its **order type**: who came before whom. And this invariant is also immune to the noise of §2 — birth *values* fluctuate in felt terms, but recorded birth *order* is an append-only fact (theory.md §6).

This closes the arc with an explanation the essay could only gesture at: **birth order is the most robust fixed gap in existence** — the invariant that survives maximal symmetry (any re-reading of time whatsoever) *and* maximal noise. Cultures disagree about calendars,

i.e. about gap values; they cannot disagree about order. That the most institutionalized kinship invariants across human societies — seniority, primogeniture, elder/younger kin terms — are precisely the ordinal ones is what this theorem would predict: institutions crystallize around the invariants that require the least shared measurement to verify.

4. State of the theory

question	answer	where
Are all recurrent readings calendar-like?	No — each one-parameter group has its own anniversary theory; a.p. iff drift-driven	§1
Does the fading law survive noise?	In form yes; but noise dominates past $t^* \asymp (\Delta d)^2/\sigma^2$, and only affine readings stay E-fixed	§2
What freezes for n objects, and how far does the invariant tower go?	Functions of the gap matrix; the tower gap $\rightarrow$ ratio $\rightarrow$ cross-ratio terminates (Tits), leaving order as the final invariant	§3

Remaining genuinely open — *all three resolved in forces.md*:

1. **Heavy-tailed noise.** Thm 2.4's horizon assumed finite variance; under α -stable idiosyncratic shocks the felt-order horizon should scale as $(\Delta d)^\alpha/\sigma^\alpha$ — is order *more* or less durable in a world of rare large shocks? **Resolved (less, in every sense):** horizon $(\Delta d)^\alpha$, no exponentially safe period, inversions caused by single identifiable jumps; below $\alpha = 1$ the fading law itself disintegrates (forces.md §1).
2. **Interacting drifts.** Objects whose drift rates depend on their gaps (attraction/repulsion): when do fixed gaps survive as relative equilibria, and is there a Lyapunov version of the rigidity theorem? **Resolved:** fixed gaps become critical points of a potential, durable iff minima (Lyapunov rigidity); defended gaps fluctuate but never dissolve under noise, reversing §2's verdict — maintenance joins ritual and records (forces.md §2).
3. **Higher dimensions.** On $\mathbb{R}^n$ or curved spaces the sharp transitivity ladder differs (isometries, conformal maps); compute the fundamental-invariant tower for the conformal group — the angle should appear where the ratio did. **Resolved (confirmed and sharpened):** gap vector $\rightarrow$ shape/angle $\rightarrow$ cross-ratio, terminating by Liouville rigidity; at maximal symmetry the surviving invariants are the components of configuration space — order on the line, cyclic order on the circle, nothing in dimension ≥ 2 . Birth order is unkillable only because time is one-dimensional (forces.md §3).

Forces, Black Swans, and the Privilege of the Line

Resolutions of the three remaining problems of frontier.md §4.

Summary of verdicts:

- **Q1 (heavy-tailed noise).** Order is *less* durable under rare large shocks, at every horizon: the felt-order horizon scales as $(\Delta d)^\alpha$ (Thm 1.1), no relationship is ever exponentially safe, and inversions are caused by single identifiable shocks rather than anonymous erosion (§1.3). Below $\alpha = 1$ the fading law itself disintegrates: at $\alpha = 1$ fading acquires a *random destination*, and for $\alpha < 1$ noise outruns time itself (Thm 1.4).
- **Q2 (interacting drifts).** When drift depends on the gaps, rigidity fails and is replaced by a Lyapunov theory: fixed gaps are the relative equilibria of a gradient flow, durable iff energy minima (Thm 2.2). This introduces a second mechanism of permanence — **defended** gaps rather than **inertial** ones — and it reverses the noise verdict of frontier.md: defended gaps fluctuate but never dissolve (Thm 2.4). Maintenance joins ritual and records as the third remedy against decay.
- **Q3 (higher dimensions).** The invariant ladder generalizes — gap vector, then *shape/angle* (where the ratio stood), then cross-ratio — and terminates by Liouville conformal rigidity (§3.2). At maximal symmetry the surviving invariants of n points are exactly the connected components of configuration space (Thm 3.3): orderings on the line, cyclic orders on the circle, **nothing at all** in dimension ≥ 2 . Birth order is unkillable because time is one-dimensional.

1. Heavy-tailed noise: black-swan inversions

1.1 Setup and the horizon. Replace the finite-variance shocks of frontier.md §2 with symmetric α -stable idiosyncratic shocks of scale σ , $\alpha \in (0, 2)$. By stability, the accumulated noise satisfies $S_t \stackrel{d}{=} t^{1/\alpha} S_1$: fluctuations grow like $t^{1/\alpha}$, faster than the Gaussian $\sqrt{t}$ for every $\alpha < 2$.

Theorem 1.1 (heavy-tailed felt-order horizon). For two pairs with gaps $d > d'$, the inversion probability is $p(t) = \mathbb{P}(S > \Delta d / (c \sigma t^{1/\alpha}))$ for a standard stable S , rising to $1/2$ with crossover horizon

$$t^* \asymp \left(\frac{\Delta d}{\sigma} \right)^\alpha.$$

Proof. The difference of the two pairs' accumulated noises is again symmetric α -stable of scale $\propto \sigma t^{1/\alpha}$; the inversion event compares it to Δd , and the scale matches the threshold when $t^{1/\alpha} \sigma \asymp \Delta d$. ■

Against the Gaussian horizon $(\Delta d)^2/\sigma^2$: doubling the gap difference buys $2^\alpha < 4$ times the durability. The heavier the tail, the weaker the durability premium of a large gap; in the Cauchy world ($\alpha = 1$) durability is merely *linear* in the gap.

1.2 No relationship is ever safe. At any early time, the one-step inversion probability is already polynomial, $p \sim c_\alpha (\sigma/\Delta d)^\alpha$, by the stable tail $\mathbb{P}(S_1 > x) \sim C_\alpha x^{-\alpha}$ — versus the Gaussian $\exp(-(\Delta d)^2/\sigma^2)$, which is astronomically small for large gaps. Heavy tails abolish the “safe early period”: even the widest felt gap carries non-negligible immediate inversion risk.

1.3 The single-big-jump principle. For subexponential (in particular stable) shocks, $\mathbb{P}(S_t > x) \sim t \mathbb{P}(\eta_1 > x)$ as $x \rightarrow \infty$: conditional on a large deviation, it was produced by *one* jump, with probability tending to 1. Consequently: **in Gaussian worlds, felt order dissolves gradually and anonymously, by the accumulation of many small shocks; in heavy-tailed worlds it fails suddenly and attributably, by a single identifiable event.** Erosion versus betrayal — the two phenomenologies of lost order are the two tail regimes of the same theorem.

1.4 Theorem (the fading-law trichotomy). For the ratio reading under α -stable idiosyncratic noise:

1. $\alpha \in (1, 2]$: the law of large numbers holds, the universal limit survives ($m(t) \rightarrow 1$ a.s.), and the noise term $\sim t^{-(1-1/\alpha)}$ still dominates the gap memory d/t , with $\text{SNR} \sim t^{-1/\alpha}$.
2. $\alpha = 1$ (Cauchy): $S_t/t \stackrel{d}{=} S_1$ for every t — the felt gap converges *in distribution to a nondegenerate random variable*. Fading acquires a random destination: pairs no longer all read the same in the end, and which “limit” a pair exhibits is never settled almost surely.
3. $\alpha \in (0, 1)$: $t^{1/\alpha} \gg t$ — noise outruns the drift. Potentials themselves are fluctuation-dominated; even the *sign* of felt seniority churns forever. There is no fading law because there is nothing left to fade toward: turbulence has beaten time.

Proof. (1): finite mean, SLLN, and exponent comparison $1 - 1/\alpha < 1$. (2): exact stability scaling at $\alpha = 1$; a.s. convergence fails since $\limsup |S_t|/t = \infty$. (3): scale comparison $t^{1/\alpha}/t \rightarrow \infty$. ■

So the answer to frontier.md's question — “is order more or less durable in a world of rare large shocks?” — is *less, at every horizon, in every sense*: shorter horizons, no safe period, and below $\alpha = 1$ the collapse of the fading framework itself. What heavy tails add is not durability but *attributability* (§1.3).

2. Interacting drifts: from frozen gaps to defended gaps

2.1 Setup. So far, objects drifted blindly: $\dot{x}_i = 1$. Now let the drift *respond to the gaps* — attraction, repulsion, imitation, competition:

$$\dot{x}_i = c + \sum_{j \neq i} \varphi(x_j - x_i),$$

with interaction kernel φ . For two bodies the gap $\delta = x_b - x_a$ obeys the autonomous equation

$$\dot{\delta} = \psi(\delta), \quad \psi(\delta) := \varphi(-\delta) - \varphi(\delta),$$

and ψ is automatically odd. Gaps have become *dynamical variables*.

Theorem 2.2 (Lyapunov rigidity). Consider the two-body gap flow $\dot{\delta} = \psi(\delta)$, and set $V(\delta) := -\int^\delta \psi$.

1. (*Rigidity as the force-free case.*) Every gap is fixed for every pair iff $\psi \equiv 0$ iff the drift is uniform — the rigidity theorem (theory.md 2.3) is exactly the degenerate, interaction-free instance.
2. Otherwise, the fixed gaps are the zeros of ψ (relative equilibria: the pair translates rigidly at a common speed), and V is a Lyapunov function: $\frac{d}{dt}V(\delta(t)) = -\psi(\delta)^2 \leq 0$. The *durable* fixed gaps are the strict local minima of V (where $\psi'(\delta^*) < 0$).
3. (*Many bodies.*) For a symmetric pair potential ($\varphi = W'$, W even), the n -body system is a uniformly translating gradient flow: the center of mass moves at speed c , and on the gap space $\mathbb{R}^n/\Delta$ (the same quotient as theory.md §3) the energy $\mathcal{U} = \sum_{i < j} W(x_i - x_j)$ strictly decreases off equilibria. Fixed gap configurations are the critical points of $\mathcal{U}$; durable ones are its local minima.

Proof. (1) restates rigidity. (2): chain rule gives $\dot{V} = -\psi(\delta)\dot{\delta} = -\psi^2$; linearization at a zero gives the stability criterion. (3): translation invariance of $\mathcal{U}$ makes $\sum_i \partial_i \mathcal{U} = 0$ (center of mass unaffected), and $\dot{\mathcal{U}} = \sum_i \partial_i \mathcal{U} \dot{x}_i = -\sum_i (\partial_i \mathcal{U})^2 \leq 0$. ■

2.3 The two mechanisms of permanence. Theorem 2.2 splits the world's fixed gaps into two kinds:

- **Inertial gaps** (theory.md): fixed because nothing acts on them — everyone drifts alike. Ages. Exact, effortless, but *undefended*.
- **Defended gaps** (new): fixed because forces restore them — deviations are pushed back. Crystal lattice spacings (short-range repulsion + long-range attraction gives nonzero minima of W : emergent preferred gaps), price spreads under arbitrage, homeostatic setpoints, institutional pecking orders. Approximate, costly, but *maintained*.

Pure attraction ($\psi' < 0$ everywhere, e.g. $\psi = -2k\delta$ from imitation dynamics) has only one equilibrium: $\delta^* = 0$. Gaps close exponentially, $\delta(t) = \delta e^{-2kt}$ — note this is a *different decay law* from the hyperbolic fading of readings.md: interaction kills the gap itself, not merely its felt size. Conformity-driven worlds (opinion dynamics, consensus processes) are exactly this regime: the only fixed gap they tolerate is zero. Repulsion at short range makes $\delta = 0$ unstable and creates nonzero stable gaps: differentiation, spacing, niche separation.

2.4 Theorem (interaction reverses the noise verdict). Add idiosyncratic noise of intensity σ (as in frontier.md §2) to the two-body flow near a stable gap δ^* with restoring rate $k := -\psi'(\delta^*) > 0$: $d\delta = \psi(\delta) dt + \sqrt{2}\sigma dB$. Then:

1. **Inertial gaps** ($\psi \equiv 0$) dissolve: variance $2\sigma^2 t \rightarrow \infty$ (frontier.md 2.2).

2. **Defended gaps** persist: the linearized gap is an Ornstein–Uhlenbeck process with *stationary* distribution $\mathcal{N}(\delta^*, \sigma^2/k)$ — fluctuation bounded uniformly in time. Felt order between defended gaps separated by more than a few stationary standard deviations retains a *permanent* positive signal-to-noise ratio: the felt-order horizon of frontier.md 2.4 becomes infinite.
3. Defended gaps fail differently: not by diffusion but by *escape* from the basin of V . Under Gaussian noise the mean escape time is exponentially long, $\asymp e^{\Delta V/\sigma^2}$ (Kramers): long quiescence, then a rapid transition to another equilibrium — punctuated equilibrium as a corollary. Under α -stable noise, escape times are only *polynomial* in $1/\sigma$ and occur by a single big jump (Imkeller–Pavlyukevich), reconnecting with §1: heavy tails convert even defended order from “practically eternal” to “merely long-lived, and lost in one identifiable blow.”

Proof sketch. (1) is frontier.md 2.2. (2): linearize; OU stationary variance is (noise variance rate)/(2 · restoring rate) = $2\sigma^2/(2k) = \sigma^2/k$; order comparison as in frontier.md 2.4 but with t -independent variances. (3): classical Kramers / Lévy-flight exit asymptotics. ■

2.5 The maintenance triptych becomes a quartet. Frontier.md §2.6 ended with: append-only ontology makes gaps *fixed*, ritual keeps them *felt*, records keep them *known*. Interaction adds the fourth and most active remedy: **restoring forces keep gaps true** — not by freezing them but by defending them. And the theory now explains a design choice visible everywhere in institutions: anything that must survive a noisy world cannot rely on inertia (Thm 2.4.1); it must either be written down (records), be periodically re-enacted (ritual), or be actively defended by a restoring force (maintenance, enforcement, homeostasis). Constitutions use all three.

3. Higher dimensions: the conformal ladder and the privilege of the line

3.1 The ladder in $\mathbb{R}^n$. Frontier.md §3 built the tower gap $\rightarrow$ ratio $\rightarrow$ cross-ratio on the line via sharp transitivity. In $\mathbb{R}^n$ the same normalize-and-read-off construction gives, group by group:

group of $\mathbb{R}^n$ (or S^n)	first nontrivial arity	fundamental invariant
translations	2	the gap vector $v - u$
similarities (isometries + scaling)	3	the shape of the triangle — its angles
conformal / Möbius group	4	the absolute cross-ratio $\frac{ x_1 - x_3 x_2 - x_4 }{ x_1 - x_4 x_2 - x_3 }$

The prediction of frontier.md §4.3 is confirmed and sharpened: *the angle is the higher-dimensional ratio*. On the line, the similarity class of three points is the ratio $(w - u)/(v - u)$; in the plane, the similarity class of three points is the triangle’s angles. Same slot in the ladder, same construction (send one point to a base point, normalize the scale using the second, read off what remains of the third); the extra dimensions turn the leftover number

into a shape. Likewise the Möbius normalization $(u, v, w) \mapsto (0, 1, \infty)$ — send a point to infinity and the remaining group is the similarity group, which handles two more points — works in every dimension, putting the first conformal invariant at four points, exactly as on the line.

3.2 Termination, again. On the line the tower stopped by Tits’s theorem on sharp 4-transitivity. In dimension $n \geq 3$ it stops for a different and stronger reason: **Liouville’s conformal rigidity theorem** — every conformal map of a domain in $\mathbb{R}^{n \geq 3}$ is the restriction of a Möbius transformation. There is no larger geometry-preserving flexibility above the conformal group; the ladder is complete at the cross-ratio. (In dimension 2 the *local* conformal maps are all holomorphic functions — an infinite-dimensional pseudo-group, the loophole that complex analysis lives in — but globally, on the sphere, the group is again just Möbius and the ladder still stops.)

3.3 Theorem (the unkillable invariants are the components of configuration space). Let the full orientation-preserving homeomorphism group of a space X act on n -tuples of distinct points. The invariants that survive are exactly the locally constant ones: the **connected components of the configuration space** $\text{Conf}_n(X)$. In particular:

1. $X = \mathbb{R}$ (the timeline): $\text{Conf}_n(\mathbb{R})$ has $n!$ components — the **orderings**. Birth order survives.
2. $X = S^1$ (the calendar): the components are the **cyclic orders** — calendar order survives, but “first” does not.
3. $X = \mathbb{R}^{d \geq 2}$: $\text{Conf}_n(\mathbb{R}^d)$ is **connected** — *nothing survives*. The group acts transitively on n -tuples for every n .

Proof. An invariant is constant on orbits; orbits of a group containing all orientation-preserving homeomorphisms are unions of components of $\text{Conf}_n(X)$, and within a component any tuple can be carried to any other by an isotopy, extended to an ambient homeomorphism. Counting components: on the line, tuples are separated by their sort order ($n!$ components); on the circle, by cyclic order; for $d \geq 2$ one can move points around each other, so $\text{Conf}_n(\mathbb{R}^d)$ is path-connected. ■

3.4 The privilege of the line. Frontier.md ended by crowning birth order the most robust fixed gap in existence. Theorem 3.3 reveals *why*, and how contingent it is: **order survives maximal symmetry because — and only because — time is one-dimensional**. In a plane there is no “before”; points in dimension ≥ 2 can be carried around one another, and with that single topological fact every last invariant dies. The unkillable invariant of our lives is not a deep necessity of invariance theory; it is a low-dimensional accident — the timeline is too narrow for histories to slip past each other.

Two corollaries of the worldview kind:

- *Cyclic time keeps order but loses origins.* On a circular timeline (Thm 3.3.2), what survives is cyclic order — after/after/after, with no first. Calendrical cultures that model time as a wheel retain seniority relations but provably cannot ground “the eldest of all”: an absolute firstborn requires a line, not a loop.
- *Relativity rhymes with this rather than breaking it.* In Minkowski spacetime, one-dimensionality survives along causal directions, and order survives exactly there — as the causal partial order. The Alexandrov–Zeeman theorem states the converse jewel: any bijection of Minkowski space preserving *just* the causal order is already a Poincaré transformation up to dilation. Order is not merely what survives the geometry; in Lorentzian signature, order *generates* the geometry.

4. Where the theory now stands

The arc, document by document: gaps are frozen relations (essay); frozen means drift-invariant, and the gap table is the world up to gauge (theory); what changes is readings, in four signatures, fading hyperbolically (readings); noise, slow anniversaries, and the invariant ladder up to cross-ratio and order (frontier); and now — black swans break order attributably, forces replace inertia with defense, and order itself is the privilege of a one-dimensional time (forces).

Still genuinely open — *all three resolved in finale.md*:

1. **Defended gaps under structural change.** Thm 2.4 treats a static potential V ; let V itself drift slowly (institutions decay, forces weaken). Adiabatic theory of defended gaps: how slowly must V change for a defended gap to track its moving equilibrium, and is there a critical melting rate? **Resolved:** tracking lag = rate/stiffness; three deaths (snap, tip with an $\epsilon^{2/3}$ overshoot law, leak), all announced by rising variance — critical slowing down as a general early-warning theorem for defended invariants (finale.md §1).
2. **Networks of gaps.** n -body interactions on a graph (only some pairs interact): when does local defense of gaps imply global rigidity of the configuration? This is a fixed-gap version of rigidity percolation. **Resolved:** on the line, rigidity = connectivity; realizability of preferred gaps is exactness of a 1-cochain (frustration = cohomology, with the original gap structures as the frustration-free case), and gap fluctuation = noise $\times$ effective resistance — chains lose long-range order, expanders keep it, records are long-range edges (finale.md §2).
3. **The Lorentzian ladder.** Compute the fundamental-invariant tower for the causal group of Minkowski space (arity by arity, as in §3.1), with Alexandrov–Zeeman as the termination theorem. The conjecture: the proper time along a chain plays the gap’s role, and the tower is shorter than the conformal one. **Resolved (conjecture half right):** the tower has the same three rungs — interval, interval ratios (the twin observable), interval cross-ratios (the CFT invariants) — but the discrete bottom rung is self-sufficient: by Zeeman’s theorem, causal order alone generates the geometry in dimension ≥ 3 . Age is the longest causal chain; causal set theory supplies the closing inversion (finale.md §3).

Melting, Frustration, and the Causal Web

Resolutions of the three remaining problems of forces.md §4 — and the close of the arc.

Summary of verdicts:

- **Q1 (melting of defended gaps).** There is an adiabatic theory: a defended gap tracks its drifting equilibrium with lag proportional to (rate of change)/(stiffness), and it dies in exactly three ways — *snap* (drift too fast), *tip* (saddle-node melting, with a universal $\epsilon^{2/3}$ overshoot law), or *leak* (noise-driven escape when the barrier falls to the noise temperature). All three deaths share one observable prodrome: rising fluctuation and slowing recovery — the variance of a defended gap inflates before it dies (§1).
- **Q2 (networks of gaps).** Local defense implies global rigidity on the line iff the interaction graph is connected; the obstruction to a *consistent* network of preferred gaps is cohomological (frustration = the cycle-space component of the preferred-gap cochain), and the original gap structures of theory.md are exactly the frustration-free case. Under noise, the fluctuation of any gap equals noise intensity times **effective resistance** — so chains lose long-range order, planar networks lose it logarithmically, and only well-connected (transient, expander-like) comparison networks keep a globally rigid order (§2).
- **Q3 (the Lorentzian ladder).** The tower in Minkowski space is exactly parallel to the Euclidean one — interval (2 points) → interval ratios (3 points, the twin-paradox observable) → interval cross-ratios (4 points, the invariants of conformal field theory) — and terminates by the Lorentzian Liouville theorem. The conjecture that the tower is *shorter* was wrong; the truth is stranger: by Zeeman's theorem the discrete bottom rung (causal order alone) already generates the entire group in dimension ≥ 3 . And causal set theory proposes the final inversion: the world does not *have* a gap structure — it *is* one (§3).

1. Melting: how defended gaps die

1.1 Setup. Take a defended gap (forces.md §2) whose potential drifts slowly: $\dot{\delta} = \psi(\delta, s)$ with slow time $s = \epsilon t$, equilibrium branch $\delta^*(s)$ (where $\psi = 0$), and stiffness $k(s) = -\partial_{\delta}\psi(\delta^*(s), s) > 0$. The institution, the force, the habit that defends the gap is itself changing.

1.2 Lemma (tracking lag). Writing $\delta(t) = \delta^*(\epsilon t) + \eta(t)$ and linearizing, $\dot{\eta} = -k\eta - \epsilon \delta^{*'}(s)$, whose quasi-stationary solution is

$$\eta_{\text{qs}} = -\epsilon \frac{\delta^{*'}(s)}{k(s)}.$$

A defended gap follows its moving equilibrium with a lag equal to *(speed of the target) / (stiffness of the defense)*, scaled by the melting rate. Slow change is tracked gracefully; the tradition bends.

1.3 The three deaths. The lemma fails in three distinct ways, which exhaust the phenomenology:

1. **Snap** — the lag exceeds the basin: if $\epsilon |\delta^*|/k > R$ (the basin half-width), the gap detaches from its equilibrium and the dynamics carries it elsewhere. Critical melting rate: $\epsilon_c \approx kR/|\delta^*|$. Yes, there is a critical rate — reform faster than ϵ_c and the gap does not adapt; it breaks.
2. **Tip** — the equilibrium itself is annihilated: as V drifts, a minimum can merge with the adjacent maximum (saddle-node), $k(s) \rightarrow 0$ at some s_c . No slowness saves the gap here, but its death is *delayed*: by the classical dynamic saddle-node analysis ($\dot{x} = \mu(\epsilon t) + x^2$), the gap survives past the static melting point by a parameter overshoot of order $\epsilon^{2/3}$ — the universal lag of tipping points. Institutions outlive the conditions that sustained them, by a computable margin.
3. **Leak** — noise preempts geometry: with noise intensity σ , the Kramers escape rate $\sim e^{-2\Delta V(s)/\sigma^2}$ becomes order-one when the shrinking barrier reaches the noise temperature, $\Delta V(s) \sim \sigma^2$. The gap dies *before* the static or dynamic melting point — turbulent worlds melt their defended gaps early. (Under heavy-tailed noise, by forces.md 2.4.3, the leak is polynomial-fast and happens by one identifiable shock.)

1.4 The early-warning theorem. All three deaths are preceded by the same observable signature. Near melting, $k(s) \rightarrow 0$ (tip) or the effective barrier shrinks (leak), and the stationary fluctuation of the defended gap is σ^2/k with autocorrelation time $1/k$: **the variance of a defended gap inflates, and its recovery from perturbations slows, before it dies.** This is critical slowing down, familiar as the early-warning signal of tipping points in climate and ecosystems — here derived as a general fact about *any* defended invariant. A relationship, a norm, a peg, a spacing that starts fluctuating more and recentering slower is not “still fine but noisy”; it is announcing the death of its restoring force. Watch the variance, not the mean: the mean is the last thing to move.

2. Networks: frustration is cohomology, defense is resistance

2.1 Setup. Objects $1, \dots, n$ on the line; an interaction graph G with edge set E ; only adjacent pairs defend their gap, with preferred (signed) gaps g_e and quadratic energy $\mathcal{U}(x) = \sum_{e=(i,j) \in E} \frac{k_e}{2} ((x_j - x_i) - g_e)^2$. When does local defense pin the *global* configuration — including the gaps of non-adjacent pairs?

2.2 Theorem (connectivity is rigidity; frustration is cohomology).

1. The configuration modulo translation is determined by the edge gaps iff G is connected: gaps propagate along paths by the cocycle identity, so a spanning tree of defended edges pins every pairwise gap.

2. The preferred gaps $\{g_e\}$ are simultaneously realizable iff the 1-cochain g is **exact** ($g_e = f_j - f_i$ for node values f), iff its sum around every cycle vanishes. The obstruction lives in the cycle space $\cong H^1(G; \mathbb{R})$: each independent cycle carries a **frustration number** (the sum of preferred gaps around it).
3. In general the equilibrium realizes the (k_e -weighted) orthogonal projection of g onto the exact cochains — a discrete Hodge decomposition. The residual cycle component is never realized; it is distributed as permanent strain, with minimum energy $\frac{1}{2} \|g_{\text{cyc}}\|_k^2$.

Proof. (1): differences telescope along paths; connectivity makes every pair joined. (2): exactness $\Leftrightarrow$ vanishing cycle sums is the elementary homology of graphs. (3): minimizing a quadratic in dx over the space of exact cochains is the stated projection. ■

2.3 The origin story of the cocycle identity. Part (2) reframes the entire theory: theory.md's Definition 1.1 — a gap structure is a δ satisfying the cocycle identity — is precisely the statement that δ is an *exact cochain on the complete graph*, and Prop 1.3 (gauge freedom) is its potential. So the original theory was, from the start, the **frustration-free case** of a general theory of gap networks. Real social and physical networks need not be frustration-free: A wants a certain gap to B, B to C, and C to A, and the three preferences may not sum to zero. Then *no* configuration satisfies everyone; the equilibrium is the Hodge projection — the closest consistent world — and the frustration numbers measure, cycle by cycle, how much relational preference is structurally unsatisfiable. Some conflict is not misunderstanding; it is cohomology.

2.4 Theorem (defense is resistance). Add idiosyncratic noise of intensity σ to each node of a connected defended network (Langevin dynamics on $\mathcal{U}$). The stationary fluctuation of the gap between *any* two nodes — adjacent or not — is

$$\text{Var}(x_i - x_j) = \frac{\sigma^2}{2} R_{\text{eff}}(i, j),$$

the **effective resistance** between i and j in the network with edge conductances k_e (the stationary law is the Gaussian free field on G).

Consequences (with the classical asymptotics of R_{eff}):

1. **Chains lose order.** On a path/1-D lattice (each rank comparing only to its neighbors), $R_{\text{eff}}(i, j) = |i - j|/k$: gap variance grows *linearly* with separation. A hierarchy defended only by adjacent-rank comparisons drifts unboundedly at long range — rank inflation as the social instance of the Landau–Peierls theorem (no 1-D crystalline long-range order).
2. **Planar networks are marginal.** On a 2-D lattice, $R_{\text{eff}} \sim \log |i - j|/k$: order decays, but only logarithmically.
3. **Well-connected networks keep global order.** On transient graphs (3-D lattices) and expanders, R_{eff} is uniformly bounded: every gap in the network, however distant, fluctuates within a fixed band. Global rigidity of a defended order is a *connectivity* property — it requires comparison paths that are numerous and short.
4. **Records are long-range edges.** A written record fixing the gap between two distant nodes is an edge of enormous conductance spliced across the network, collapsing R_{eff} for every pair whose paths route through it. The epistemic role of archives (frontier.md §2.6) acquires a structural counterpart: archives are the shortcuts that make a comparison network transient.

2.5 Rigidity percolation. If each potential comparison edge exists independently with probability p , global rigidity on the line is exactly connectivity of the random graph, so the rigidity threshold is the connectivity threshold (for $G(n, p)$: $p = \log n/n$). In genuinely $d \geq 2$ -dimensional embeddings, rigidity requires more than connectivity (Laman-type counting; generic rigidity percolation has a strictly higher threshold) — pinning *shapes* is harder than pinning *lines*, consistent with §3’s theme that dimension changes what invariants cost.

3. The Lorentzian ladder: order generates geometry

3.1 The ladder. Compute the fundamental-invariant tower of Minkowski space $\mathbb{R}^{1,n-1}$ ($n \geq 3$), group by group, exactly as in forces.md §3.1:

group	first nontrivial arity	fundamental invariant
Poincaré	2	the spacetime interval $I(x, y)$ — the gap
causal group (Poincaré $\times$ dilations)	2 (discrete), 3 (continuous)	the causal type of a pair (before / after / elsewhere / lightlike); then interval ratios
Lorentzian conformal group $O(2, n)$	4	the interval cross-ratios $\frac{I_{13} I_{24}}{I_{14} I_{23}}$

Reading the rungs: the Poincaré-level invariant of a pair is the interval — the gap, as theory.md §5 already tabulated. Adding dilations (the causal group) kills the interval’s magnitude, leaving a *discrete* pair invariant — the causal relation itself — with the first continuous invariant appearing at three points as ratios of intervals: for three causally related events, the ratio of proper times along the two legs. This is precisely the twin-paradox observable — *how much more one path ages than another* — the Lorentzian “shape,” standing exactly where the angle stood in Euclidean space and the ratio stood on the line. At four points, the conformal group leaves the interval cross-ratios: exactly the variables on which conformal field theory correlators depend. The tower terminates by the Lorentzian Liouville theorem: for $n \geq 3$ the conformal transformations of Minkowski space form the finite-dimensional group $O(2, n)$; there is nothing above.

3.2 Scoring the conjecture. Forces.md §4.3 conjectured that proper time plays the gap’s role (correct, at the Poincaré rung) and that the tower is *shorter* than the conformal one (wrong). The tower has the same three rungs — gap, shape, cross-ratio. What is genuinely different in Lorentzian signature is the *bottom*:

Theorem (Zeeman, 1964). For $\dim \geq 3$, every bijection of Minkowski space preserving the causal order (in both directions) is an orthochronous Poincaré transformation composed with a dilation.

The discrete arity-2 invariant — mere order, the weakest-looking rung — already pins the entire group, hence the entire ladder above it. In Euclidean space, order was the last survivor of maximal symmetry (forces.md 3.3); in Lorentzian signature it is promoted to *generator*:

preserve who-can-influence-whom, and you have preserved the geometry, the intervals up to scale, everything. (Dimensional wrinkle, mirroring the privilege of the line: in 1+1 dimensions Zeeman fails — the causal automorphisms $(u, v) \mapsto (f(u), g(v))$ on null coordinates form an infinite-dimensional group. Order generates geometry only when spacetime is wide enough for light cones to interlock.)

3.3 Age as the longest chain. In Minkowski space the reverse triangle inequality holds: for timelike-related events, $\tau(x, z) \geq \tau(x, y) + \tau(y, z)$ — the straight path through spacetime is the *oldest*. Proper time, the relativistic age, is the length of the longest causal chain between two events. The essay’s opening quantity — age — reappears at the foundation of physics not as a coordinate but as an extremal statistic of the order itself.

3.4 The final inversion. Causal set theory (Bombelli–Lee–Meyer–Sorkin) takes the last step: it proposes that spacetime *is* a locally finite partial order — a discrete web of fixed precedence-gaps — from which geometry is recovered as bookkeeping: volume from counting elements, proper time from longest chains, dimension and curvature from order statistics. Its slogan: *order plus number equals geometry*.

If anything like this is right, the arc of these documents closes on an inversion. We began with gaps as frozen facts *between* things — ages, kinships — held in place by a world. Every document since has stripped the things of standing: only gaps are measurable (theory), objects are recoverable from gaps up to gauge (theory), what changes is only readings (readings), what survives symmetry and noise is order (frontier, forces). The causal-set endpoint is the limit of that progression: there is no world holding the gaps in place. The web of fixed gaps is the world, and spacetime, matter, and we ourselves are its readings.

4. Coda: the arc, complete

Each document’s slogan, in order:

1. **essay.md** — between people and objects alike, what persists are relations, not conditions.
2. **theory.md** — gaps are fixed iff everything drifts together; the gap table is the world up to gauge; append-only worlds fix their positive facts.
3. **readings.md** — meaning is a reading over a frozen substrate: four signatures, hyperbolic fading, and anniversaries as quotient gaps.
4. **frontier.md** — recurrence is relative to a group; noise buries felt order at a quadratic horizon; the invariant tower ends at the cross-ratio, leaving birth order.
5. **forces.md** — heavy tails break order attributably; forces replace inertia with defense; order is the privilege of a one-dimensional time.
6. **finale.md** — defended gaps die by snap, tip, or leak, announced by rising variance; frustration is cohomology and defense is resistance; and in the deepest description we have, the world is not a place where gaps happen to be fixed — it is the fixed gaps, being read.

The program as originally posed — formalize the observation that some relations never change, and see how far it goes — is complete. What remains is no longer this theory’s to settle: whether causal-set recovery of geometry can be made quantitative is working

physics; whether entanglement structure admits a gap-theoretic reading is quantum information; whether frustration numbers of real social networks predict where their conflicts concentrate is empirical sociology. The mathematics has carried the observation to the borders of three sciences, which is where an essay's idea should hope to end up.

The Worked Application: Rating Systems as Defended Gap Networks

An external review of this monograph judged its main weakness to be the absence of a “killer application” — a problem the framework explains or predicts that is not naturally reached from existing viewpoints. This chapter is the response: one domain worked end to end, from the framework’s concepts to falsifiable quantitative laws to numerical verification. The simulation code is in `simulations/ratings_sim.py`; all numbers below are from the committed run (`simulations/results.txt`).

The domain is **pairwise-comparison rating systems** — chess Elo, sports rankings, and LLM leaderboards such as Chatbot Arena — chosen because they are, without any modeling license, the objects this theory is about: a rating system is a defended gap network in the sense of `finale.md §2`, running live on real data at scale.

The chapter’s findings, in one paragraph: the framework’s central distinction — *where the noise enters* (nodes vs edges) versus *where the defense acts* (edges) — cleanly splits rating systems into three regimes with three different laws. Estimation error of a rating gap follows the resistance law exactly (P1). The stationary churn of online Elo with static skills is **graph-independent** — a surprise the framework forces you to notice and the rating literature has little reason to ask about (P2). And when true skills drift, the resistance law returns as the *tracking* error, with a computable correction: mean reversion grounds the network (P3). Each law is verified by simulation below; the third also yields a matchmaking design rule and a derivation of chess’s era-comparison problem.

1. The mapping

The Bradley–Terry model: players i have skills s_i , and i beats j with probability $\sigma(s_i - s_j)$ (σ logistic). Every piece of the monograph’s apparatus lands on a native feature of this system:

framework (chapter)	rating system
potentials, gauge freedom (<code>theory.md 1.3</code>)	ratings; only rating <i>differences</i> predict anything — the anchor and scale (Elo’s “400 points”, the initial 1500) are pure convention, which is why ratings are notoriously incomparable across pools
gap structure on the complete graph	the true skill differences $s_i - s_j$

framework (chapter)	rating system
defended gaps, restoring force (forces.md §2)	matches: each game pulls the rating gap toward the observed outcome — Elo’s update is stochastic gradient descent on the Bradley–Terry log-likelihood
the comparison network (finale.md §2)	the match graph: who has actually played whom, with edge weights = match counts
idiosyncratic node noise (frontier.md §2)	true skill drift: players improve and decline individually
edge noise	outcome randomness: a match is a coin flip biased by the gap

The dynamics, linearized about small gaps (details in §2): with n players, match graph Laplacian L , matches arriving uniformly on $|E|$ edges, and update factor K ,

$$dr = a L (s - r) dt + \text{edge noise}, \quad a = \frac{K}{4|E|}, \quad \text{Cov}(\text{edge noise}) = \frac{K^2}{4|E|} L.$$

The structural fact that drives everything: **the noise covariance and the restoring force are built from the same Laplacian**, because in a rating system every observation is also a defense — you cannot watch a gap without tugging on it.

2. Three laws

P1 (estimation: the resistance law). With m matches per edge and static skills, the Fisher information of the Bradley–Terry model at equal skills is $(m/4)L$, so the MLE of any rating gap — between *any* two players, whether or not they ever met — has

$$\text{Var}(\widehat{r_i - r_j}) = \frac{4}{m} R_{\text{eff}}(i, j),$$

the effective resistance in the match graph. The Fisher computation is classical; the point is the identification: this is finale.md’s resistance law ($\text{Var} = (\sigma^2/2)R_{\text{eff}}$), reached here through statistics rather than Langevin dynamics — the two derivations are the same Laplacian inverse wearing different clothes. Uncertainty about a gap is a property of the *network between* the pair, not of the pair.

P2 (dynamics with static skills: the isotropy surprise). Run online Elo forever with static, equal skills. The stationary covariance solves the Lyapunov equation $2aL\Sigma = \frac{K^2}{4|E|}L$, giving $\Sigma = \frac{K}{2}P$ (P the centering projector), hence

$$\text{Var}(r_i - r_j) = K \quad \text{for every pair, on every connected graph.}$$

The resistance dependence *cancels*: because noise and defense enter through the same edges, poorly connected pairs get less restoring force but also less noise, in exactly compensating

amounts. Online Elo’s stationary churn is graph-independent — a path and a complete graph produce identical long-run gap fluctuation. We are not aware of this being stated in the rating literature; it is invisible unless one asks the framework’s question (*where does the noise enter relative to the defense?*), and it sharpens frontier.md’s noise dichotomy into a trichotomy: common noise (gaps exact), node noise (resistance law), edge noise (isotropy).

P3 (dynamics with drifting skills: the tracking resistance law). Now let true skills drift idiosyncratically — node noise of strength σ_d , the frontier.md §2 regime. The tracking error $e = r - s$ receives isotropic node noise but only edge defense, and the mode-by-mode stationary solution gives

$$\text{Var}(e_i - e_j) = \underbrace{K}_{\text{edge part (P2)}} + \underbrace{\frac{2|E|\sigma_d^2}{K} R_{\text{eff}}(i, j)}_{\text{node part: the resistance law}}.$$

Refinement: if the drift is weakly mean-reverting with rate θ (necessary for a stationary regime — see P4), the node noise is high-pass filtered and the exact linear theory replaces R_{eff} by the **grounded resistance** — the resistance of the network in which every node additionally leaks to ground through conductance θ/a , i.e. the quadratic form of $(L + (\theta/a)I)^{-1}$. Mean reversion is literally a grounding wire on every player.

P4 (saturation: the era catastrophe). If skills random-walk *without* mean reversion, skill gaps grow without bound, the logistic saturates, $\sigma'(s_i - s_j) \rightarrow 0$, and Elo’s restoring signal collapses exponentially: tracking error diverges on every graph. A bounded-update rating system cannot follow unboundedly diverging skills at all — the strongest form of the era problem.

3. Numerical verification

All from one committed run of `ratings_sim.py` ($n = 12$ – 16 players; graphs: path, cycle, two clusters joined by two bridges, random 3-regular, complete; $K = 0.1$):

P1 — regress empirical MLE gap-error variance (600 trials, 40 matches per edge) on $(4/m)R_{\text{eff}}$, all pairs:

graph	slope (theory 1)	R^2
path	1.006	0.992
cycle	1.024	0.964
two-cluster	1.020	0.994
3-regular	1.027	0.927
complete	1.017	— (all resistances equal; no variance to explain)

P2 — online Elo, static skills, 4M steps: mean stationary gap variance 0.1055–0.1060 against the predicted $K = 0.1$ (the few-percent excess is the expected nonlinear correction), and *flat in resistance*: on the path, the low-resistance half of pairs shows 0.1056 and the high-resistance half 0.1064 — adjacent and distant pairs churn identically, as predicted and contrary to the naive guess.

P3 — drifting skills ($\sigma_d = 0.004$, $\theta = 10^{-4}$, 8M steps): gap tracking-error variance is linear in R_{eff} (path $R^2 = 0.95$, two-cluster $R^2 = 0.90$) with intercept 0.106 / 0.105 against the predicted $K = 0.1$. Measured slopes: path 0.00206 vs exact grounded theory 0.00213 (3%; the naive ungrounded value 0.00352 is clearly rejected), two-cluster 0.00964 vs 0.00812 (within 20%; the grounded correction accounts for most of the gap to the naive 0.01024).

P4 — with $\theta = 0$ the path-graph tracking variance explodes (≈ 6.3 at the same parameters, sixty times the defended level, still growing), confirming saturation escape.

Design corollary — in the two-cluster graph, adding *one* cross-cluster match reduces the worst-pair resistance from 1.000 to 0.778; adding one more within-cluster match reduces it to 0.995. Per match, the bridge is $\sim 44\times$ more informative about the rankings that matter.

4. Predictions for real systems

1. **Chess’s era problem is a theorem, not an accident.** The temporal comparison graph of chess is a chain: players of 1880 never played players of 1990, so the match graph across eras is path-like and R_{eff} grows linearly with era distance. By P3, cross-era rating gaps drift with variance linear in the separation — rating inflation and the impossibility of comparing Morphy with Carlsen are finale.md’s Landau–Peierls chain instability, realized in data everyone already has. The known remedies map exactly to the theory’s: anchoring to a fixed reference (an engine, a rule) is grounding; longitudinal careers spanning eras are long-range edges.
 2. **Leaderboard confidence intervals should be resistance-based.** By P1, the uncertainty of “model A vs model B” on an arena-style leaderboard is ($4/m$ -weighted) effective resistance between them in the battle graph — not a function of how many battles each has fought. Two models each with thousands of battles but no common opponents can have an arbitrarily uncertain relative rank. This is testable against bootstrap CIs on public battle data: the framework predicts bootstrap variance of rating gaps is proportional to match-graph resistance.
 3. **Resistance-optimal matchmaking.** The design corollary generalizes: to sharpen a leaderboard fastest, schedule the comparison that most reduces the (grounded) resistance of the pairs you care about — in practice, bridges between weakly connected clusters (new vs established models, different leagues, different eras) dominate yet another match between household names. Rating systems that matchmake for entertainment systematically overspend on low-resistance edges.
 4. **A falsifiable oddity.** By P2, the long-run *churn* of a constant- K Elo pool with stable skills should show gap fluctuation $\approx K$ (in logistic units) for all pairs, independent of the match graph — measurable in any stable online pool, and a direct test of the edge-noise isotropy that the framework predicts and intuition does not.
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5. What is new here, honestly

The Bradley–Terry Fisher information and Elo-as-SGD are classical; effective resistance as statistical leverage appears across network statistics. The contributions specific to this chapter are: (i) the **noise-location trichotomy** (common / node / edge) as the organizing question for rating dynamics, and its consequence P2 — the graph-independence of stationary Elo churn — which we have not found stated elsewhere; (ii) the **grounded resistance law** P3 for tracking error under mean-reverting drift, quantitatively verified; (iii) the identification of the era problem, anchoring, and matchmaking design as instances of the monograph’s chain instability, grounding, and bridge theorems — the same mathematics that governed crystals and kinship hierarchies in `finale.md`, now producing numbers in a domain with abundant public data.

Limitations, plainly: the verification is simulation, not yet real data; the derivations are linearizations (the few-percent excesses in §3 are the visible nonlinear corrections); and prediction 2 awaits a run against actual leaderboard battle logs, which is the natural next step and requires only public datasets and the estimator already implemented in `ratings_sim.py`.

Appendix: An External Review, and a Response

In July 2026 the monograph received an external review (produced by GPT, solicited by the author). The full text is preserved below, followed by a response. The review prompted the addition of application.md.

Summary of the review

The review scores the work 8.5/10 overall as a mathematics paper (originality 9/10, technical depth 7.5/10, presentation 9.5/10), and makes five substantive claims:

1. **The early mathematics is elegant repackaging, not new machinery** — gap structures are torsors, Prop 1.3 is potentials-modulo-constants, Thm 2.3 is a familiar rigidity fact.
2. **The Reading framework is the paper’s real contribution** — the substrate/interpretation split and the four-signature classification constitute “a genuine theoretical architecture rather than merely another mathematical definition.”
3. **The conceptual significance is an invariant-first ontology** (Relations $\rightarrow$ Objects), placing the work in the tradition of Leibniz and structural realism, with better publication prospects in mathematical philosophy than in mainstream mathematics.
4. **Correctness:** “the theory appears internally consistent, and I did not notice any obvious logical flaws,” with the caveat that formal peer review would require expert verification of each theorem.
5. **The main weakness is the lack of a “killer application”** — a problem the framework uniquely explains or solves; testable predictions or explanatory power in AI, networks, or economics would substantially raise its significance.

Response

On claim 1 — accepted without reservation. The monograph says as much of itself (theory.md calls Prop 6.2 “the finger-exercise version” of preservation theorems). The early chapters buy a shared language, not new machinery; the price of admission for the later ones.

On claim 2 — agreed, and it is a perceptive identification. The substrate/reading split is the load-bearing idea: everything after readings.md (noise horizons, defended gaps, the invariant tower) is an answer to “what happens to readings when you change the substrate’s symmetry, noise, or interaction structure?”

On claim 4 — the review under-delivers here, and the point matters. “Appears internally consistent” was written about a text that contained a real quantitative error at the

time of an earlier draft: the stationary variance of a defended gap was stated as $\sigma^2/(2k)$ where the correct value under the stated noise normalization is σ^2/k . The error was found by an explicit re-derivation pass (commit 308bc02), not by the review. Impressionistic correctness assessments of mathematical texts have limited value; this one certified a document that was wrong at the moment of certification. The lesson is recorded here deliberately: verification is a computation, not a reading.

On claim 5 — the criticism was fair, and it has now been answered concretely rather than programmatically. The review asked for testable predictions or explanatory power in AI, network science, or economics. Two responses:

1. *The predictions already existed in chapters 5–6*, which the review engaged with more thinly than the early chapters: the felt-order horizon $t^* \asymp (\Delta d)^2/\sigma^2$, the early-warning law (variance inflation precedes the death of any defended gap), and the resistance law ($\text{Var} = (\sigma^2/2)R_{\text{eff}}$) are quantitative and falsifiable as stated.
2. *One application has now been worked end to end* in application.md: rating systems (chess Elo, LLM leaderboards) as defended gap networks. The framework’s noise-location question produced one prediction we have not found in the rating literature (the graph-independence of stationary Elo churn), one quantitative law verified to within a few percent in simulation (the grounded-resistance tracking law), a derivation of chess’s era problem from the chain-instability theorem, and a matchmaking design rule (bridge comparisons dominate, $\sim 44\times$ in the test configuration). The remaining step the review could legitimately still demand — a run against real leaderboard battle logs — is scoped there as next work.

On the scores — no dispute worth recording, with one observation: the review’s most valuable sentence is not a score but the diagnosis that the work’s contribution type is *conceptual unification*, and that such work is judged by whether the unification eventually pays for itself in problems solved. Application.md is the first installment of that payment; whether it suffices is for the next reviewer.